

SMART COURT BOOKING & ACCESS SYSTEM

Complete Project Documentation (Final Proposal)

1. Core System Overview

Smart Court Booking & Access System

Fully automated flow: **Booking** → **Payment** → **Access** → **Session** → **Exit**

Works with:

- Web Application (Users)
 - PWA Dashboard (Admins)
 - IoT Devices (Gate, Lighting, Router Control)
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2. User (Player) System

Booking

- View real-time court availability
- Select date & time slot
- Prevent overlapping bookings

Payments

- Online payment required before confirmation
- Auto-confirm booking after payment

Access Code

- Unique session-based access code generated
- Sent via SMS or Email
- Valid only during session + 30-minute grace period

Notifications

- Booking confirmation
- Session reminder (15 min before)
- Overstay warning



3. Admin / Controller Dashboard (PWA)



Monitoring

- Active sessions (live)
- Upcoming bookings
- System status (Gate, Lights, Router)



Controls

- Open/close gate manually
- Override lights ON/OFF
- Restart router remotely



Management

- User block/allow
- Booking edit/cancel
- Session override control



Analytics

- Revenue tracking
- Peak usage times
- Usage reports



4. Backend System (Cloud)



API Layer

- REST APIs + WebSockets
- Secure HTTPS communication
- JWT Authentication



Core Services

- Booking Service
- Payment Service
- Access Code Service
- Notification Service
- Device Control Service
- Analytics Service

Database

Stores:

- Users
- Bookings
- Payments
- Access codes
- Logs
- Devices

Real-Time Engine

- WebSockets / MQTT
 - Live session updates
 - Device status updates
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5. IoT / Hardware Layer

Edge Controllers

- Raspberry Pi 4 (Main system brain)
- ESP32 WROOM-32U (Gate controller)

Gate System

- Dahua ASR2101A IP66 Keypad Reader
- Code verification via ESP32
- Relay-based lock control
- Manual key override (emergency)

Lighting System

- Existing smart switches
- Controlled by Raspberry Pi

Sensors

- BH1750 Light Sensor (lux measurement)
- Optional: gate status / motion sensors

Router Control

- Relay-based restart system

- Ping-based failure detection
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6. Security Layer

- Secure HTTPS communication
 - Encrypted data transfer
 - Expiry-based access codes
 - Limited retry attempts (anti brute-force)
 - Role-based access control
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7. Automation & Smart Logic



Lighting Control

- Based on session schedule + ambient light (BH1750)



Gate Access

- Active only during valid session window
 - Prevent double bookings
 - Automatic session lifecycle handling
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8. Reliability & Fail-Safe

- Offline mode (cached access at gate)
 - Manual physical key override
 - Emergency exit always available
 - Automatic router restart recovery
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9. Notifications System

- Booking confirmation
 - Payment confirmation
 - Session reminders
 - Overstay alerts
 - Admin system alerts
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10. Logging & Audit

- Entry/exit logs
 - Booking history
 - Payment logs
 - Admin actions
 - Device activity logs
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11. Monitoring & Analytics

System Health

- Gate status
- Lights status
- Router connectivity

Usage Analytics

- Peak hours
 - Court utilization
 - Revenue reports
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12. SYSTEM FLOW (REAL EXAMPLE)



Scenario: Kamal books 6:00–7:00 PM slot

1. Booking

- Kamal selects time slot
- Completes online payment
- Booking confirmed

2. Access Code Generation

- System generates code: **482915**
- Sent via SMS/Email
- Valid: 6:00–7:30 PM

3. System Preparation

- Raspberry Pi loads session data
- MQTT updates ESP32 with valid code

- BH1750 checks light condition

4. Arrival at Gate

- Kamal enters code at keypad

5. Validation

- ESP32 compares input with valid session code
- If correct → approve

6. Gate Access

- Relay triggers lock release
- Gate opens

7. Lighting Control

- BH1750 + schedule determines lighting
- Lights auto ON/OFF as needed

8. Session Active

- Court used for 1 hour
- System logs activity

9. Grace Period

- 30 minutes extra access window

10. Session End

- Lights OFF automatically
- Session closed and logged



FLOW SUMMARY

Booking → Payment → Code Generation → Arrival → Keypad Entry → Validation → Gate Open → Lights Control → Session → Grace Period → Completion



13. PROJECT SCOPE SUMMARY

This system delivers a fully automated smart sports court management platform integrating booking, payments, access control, and environmental automation using IoT hardware and cloud-based services.



14. HARDWARE LIST (FINAL)



Controllers

- Raspberry Pi 4 Model B → System brain (lighting + router + logic)
- ESP32 WROOM-32U → Gate controller



Gate System

- Dahua ASR2101A IP66 Keypad Reader → Access input
- 600lbs Magnetic Lock → Primary lock (fail-safe)
- 12V Electric Strike Lock → Alternative option
- 5V 1-Channel Relay Module (x2) → Gate + router control



Power System

- 12V DC 5A Power Supply → Main system power
- LM2596 Buck Converter → 12V to 5V conversion



Sensors

- BH1750 Light Sensor → Ambient light detection



Enclosures & Installation

- IP65/IP66 Outdoor Enclosure Boxes (x2)
- Acrylic window panel (sensor protection)
- Cable glands (waterproof entry)
- Mounting brackets (ZL bracket for maglock)



Wiring & Safety

- Terminal blocks
- Shielded wiring
- Heat shrink tubing
- Fuses (12V protection)



FINAL RESULT

✓ Industrial-grade architecture ✓ Fully automated smart booking system ✓ Scalable IoT-based design ✓
Production-ready proposal document ✓ Clear separation of system layers